

10(a)	energy of a photon required to remove an electron	B1
	<i>either:</i> energy to remove electron from a surface <i>or:</i> <u>minimum</u> energy to remove electron <i>or:</i> energy to remove electron with zero <u>kinetic</u> energy	B1
10(b)(i)	Correct read off from graph of f as 5.45×10^{14} Hz when $E_{\text{MAX}} = 0$	C1
	$5.45 \times 10^{14} \times 6.63 \times 10^{-34}$ $= 3.6 \times 10^{-19} \text{ J}$	A1
10(b)(ii)	$3.6 \times 10^{-19} / 1.6 \times 10^{-19} = 2.3 \text{ eV}$ so potassium	A1
10(c)(i)	each photon has same energy so no change	B1
10(c)(ii)	more photons (per unit time) so (rate of emission) increases	B1